

KARMANYA RAINA

Mumbai | 8355864166 | rainakarmanya7@gmail.com



EDUCATION

College of Engineering Pune (COEP)

B.Tech Electrical Engineering (2024–2028) | CGPA: 7.00

Atomic Energy Central School–4, Mumbai — CBSE Class X: 87.6%

PACE Jr. College, Nerul — Maharashtra Board, Class XII: 74.17%

EXPERIENCE

NPCIL Summer Intern | June–July 2026

- Developed a MATLAB-based RTD signal filtering framework using Butterworth IIR filters.
- Tuned filter order, cutoff frequency, and sampling frequency for optimal noise rejection on RTD resistance data.
- Applied zero-phase filtering (filtfilt) and validated performance using RMSE and tolerance-based metrics.
- Completed introductory FPGA logic design exercises, including counters and shifters.

PROJECTS

PID Control System Simulations | MATLAB, Simulink | (github.com/karmanya0712/PID-Control-System-Simulations)

Designed and analyzed PID controllers for second-order systems and a DC motor model; evaluated overshoot, settling time, damping ratio, and controller gain tuning.

Digital Lock using Logic Gates

Designed a combinational digital lock circuit using basic logic gates.

TECHNICAL SKILLS

Programming: MATLAB, Python (Basic)

Engineering Software: MATLAB, Simulink

Core Areas: Digital Signal Processing, Control Systems, Signal Filtering, FPGA Fundamentals (counters, shifters)

CERTIFICATIONS

- NVIDIA Deep Learning Institute — Getting Started with Deep Learning
- MathWorks — MATLAB Onramp

ACHIEVEMENTS

Represented India at Junior World Cup Bridge (France, 2016)